

Passive Footprinting Tools - Detailed Guide

1. Whois

- **Purpose:** Gets domain registration details like registrar, email, contact, address, and name servers.
- **How to Use:**
 - **Command Line:**
whois example.com
 - **Online Tool:** <https://whois.domaintools.com>
- **Information Gathered:**
 - Domain creation/expiry date
 - Owner and admin contact
 - Registrar and nameservers

2. Nslookup (limited passive)

- **Purpose:** Queries DNS records (A, MX, NS, CNAME) for a domain.
- **How to Use:**
nslookup example.com
nslookup -type=mx example.com
- **Note:** Passive only when querying public DNS resolvers.

3. theHarvester

- **Purpose:** Collects emails, hosts, subdomains from public sources like Google, Bing, LinkedIn.
- **How to Use (Linux/Kali):**
theHarvester -d example.com -b google
- **Output:**
 - List of discovered emails
 - Subdomains and IPs

4. Recon-ng

- **Purpose:** OSINT framework that automates info gathering (emails, hosts, domains).
- **How to Use:**
recon-ng
> marketplace install recon/domains-hosts/bing_domain_web
> use recon/domains-hosts/bing_domain_web
> set SOURCE example.com
> run
- **Output:** Subdomains, emails, DNS info in module format.

5. Maltego

- **Purpose:** Visual link analysis for people, domains, companies.
- **How to Use:**
 - Download from <https://www.paterva.com>
 - Create a graph, input domain or email
 - Add transforms to get results (e.g., Whois, email search)
- **Output:** Visual graphs of connected data points

6. Google Dorking

- **Purpose:** Uses advanced Google search operators to find sensitive data.

- **How to Use:**
 - In Google Search:
site:example.com intitle:"index of"
filetype:xls site:example.com
inurl:admin site:example.com
- **Output:** Open directories, login pages, leaked documents

7. Shodan

- **Purpose:** Search engine for Internet-connected devices (e.g., routers, webcams).
- **How to Use:**
 - Website: <https://www.shodan.io>
 - Example Query:
hostname:"example.com"
port:22 country:"IN"
- **Output:** IPs, open ports, services, vulnerabilities

8. Censys

- **Purpose:** Similar to Shodan; helps find and analyze internet-facing assets.
- **How to Use:**
 - Website: <https://search.censys.io>
 - API available for scripting
- **Output:** TLS certs, server versions, IP details

9. crt.sh

- **Purpose:** Looks up SSL/TLS certificates issued to a domain.
- **How to Use:**
 - Website: <https://crt.sh>
 - Search by domain: example.com
- **Output:** List of subdomains from certificate history

10. LinkedIn / Facebook (Manual OSINT)

- **Purpose:** Discover employee names, roles, emails, and internal structure.
- **How to Use:**
 - Search: site:linkedin.com company example.com
 - Extract usernames, titles, email patterns

11. Netcraft

- **Purpose:** Shows hosting history, DNS info, and technologies used by websites.
- **How to Use:**
 - Website: <https://www.netcraft.com>
- **Output:** Server type, OS, Uptime history, Subdomains

12. DNSDumpster

- **Purpose:** Subdomain and DNS record discovery.
- **How to Use:**
 - Website: <https://dnsdumpster.com>
 - Enter domain and submit
- **Output:** Subdomains, MX, TXT records, A records

13. Email Hunter (Hunter.io)

- **Purpose:** Finds company email patterns and specific addresses.
- **How to Use:**
 - Website: <https://hunter.io>

- Enter domain to search
- **Output:** Emails with sources and verification status

14. Wayback Machine (archive.org)

- **Purpose:** View archived versions of websites.
- **How to Use:**
 - Website: <https://archive.org/web>
 - Search domain like example.com
- **Output:** Old site structure, previously exposed directories

15. HaveBeenPwned

- **Purpose:** Checks if an email or domain has been leaked in past data breaches.
- **How to Use:**
 - Website: <https://haveibeenpwned.com>
 - Search using email or domain
- **Output:** Breach source, leaked data type